Introduction to Google Colab

I learned that Google Colab is a cloud-based platform provided by Google where users can write and execute Python code directly in the browser without installing any software. It is mainly used for Python programming, data analysis, machine learning, deep learning, and data visualization.

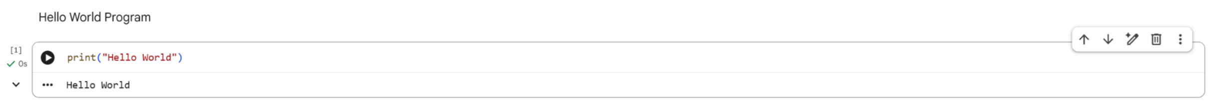
Features of Google Colab:
One important thing I learned is that Google Colab does not require installation. Everything works online, making it beginner-friendly. I also learned that Colab provides free GPU and TPU support, which helps in running machine learning and deep learning models faster.

Notebook Interface:
I got to know that the notebook interface in Google Colab. I learned about:
- Code cells
- Text or Markdown cells
- Runtime options
- Running code using Shift + Enter

Python Programming in Colab:
I learnt that how to write and execute Python programs in Colab. I learned basic coding concepts such as:
- Printing output
- Using variables
- Running Python programs step by step

Code:
print("Hello World")

Output:



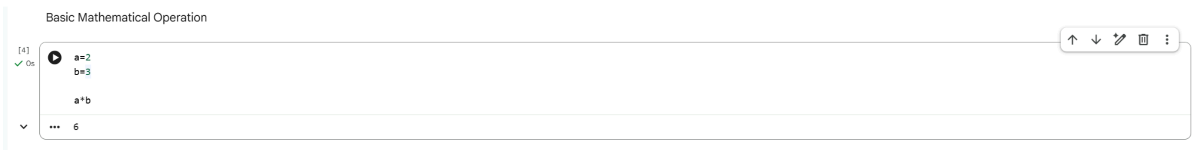
Basic Mathematical Problems:
Code:

a=2

b=3

a*b

Output:



Working With Datasets:
A major concept covered in the video was dataset handling. I learned how to:
- Upload CSV files
- Read datasets using Pandas
- View dataset records
- Understand rows and columns

Code:
import pandas as pd
df = pd.read_csv("/content/drive/MyDrive/DataSets/customers-100000.csv")
print(df.head())

Output:

```
import pandas as pd
df = pd.read_csv("/content/drive/MyDrive/DataSets/customers-100000.csv")
print(df.head())
```

```
...   Index      Customer Id First Name   Last Name                    Company  \
0      1  ffeCAb7AbcB0f07      Jared      Jarvis            Sanchez-Fletcher
1      2  b687FfC4F1600eC      Marie      Malone                   Mckay PLC
2      3  9FF9ACbc69dcF9c     Elijah     Barrera              Marks and Sons
3      4  b49edDB1295FF6E     Sheryl  Montgomery  Kirby, Vaughn and Sanders
4      5  3dcCbFEB17CCf2E     Jeremy     Houston              Lester-Manning

            City                                      Country  \
0   Hatfieldshire                                      Eritrea
1  Robertsonburgh                                     Botswana
2         Kimbury                                     Barbados
3     Briannaview  Antarctica (the territory South of 60 deg S)
4   South Brianna                                   Micronesia

               Phone 1               Phone 2  \
0    274.188.8773x41185  001-215-760-4642x969
1          283-236-9529  (189)129-8356x63741
2            8252703789    459-916-7241x0909
3          425.475.3586        (392)819-9063
4  +1-223-666-5313x4530     252-488-3850x692

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                     Website
0    https://www.mccarthy.info/
1     http://www.reynolds.com/
2            https://neal.com/
3  https://www.powell-bryan.com/
4      https://www.carrillo.com/
```

Python Libraries:
I got to know about the important Python libraries used in data science.

NumPy:
Used for numerical calculations and arrays.
import numpy as np

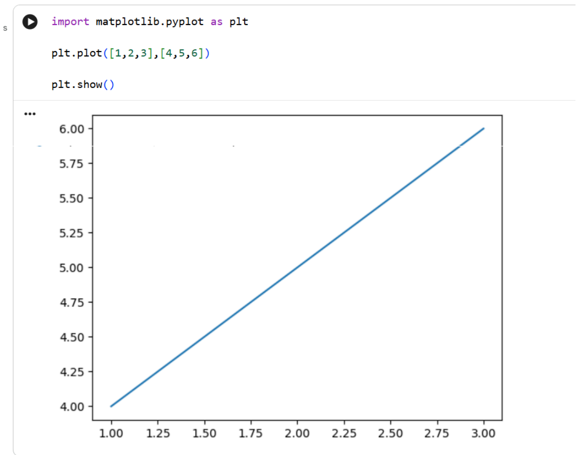
Pandas:
Used for data handling and analysis.
import pandas as pd

Matplotlib:
Used for graphs and data visualization.
import matplotlib.pyplot as plt

Data Visualization:
I learned how to create charts and graphs for visualizing data. The tutorial explained:
- Scatter plots
- Histograms
- Line charts
- Data comparison graphs

Code:
```
import matplotlib.pyplot as plt
plt.plot([1,2,3],[4,5,6])
plt.show()
```

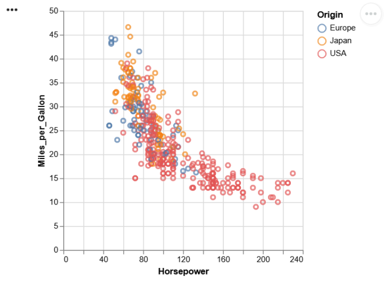
Output:



An example code of Data Visualization by loading an available online dataset by loading it and by running it:
Code:
```
from vega_datasets import data
cars = data.cars()

import altair as alt

interval = alt.selection_interval()
chart = alt.Chart(cars).mark_point().encode(
    x='Horsepower',
    y='Miles_per_Gallon',
    color=alt.condition(
        interval,
        'Origin',
        alt.value('lightgray')
    )
).add_params(
    interval
)
chart
```

Output:

```python
interval = alt.selection_interval()

chart = alt.Chart(cars).mark_point().encode(
    x='Horsepower',
    y='Miles_per_Gallon',
    color=alt.condition(
        interval,
        'Origin',
        alt.value('lightgray')
    )
).add_params(
    interval
)

chart
```



Google Drive Integration:
Another important topic I learned was connecting Google Drive with Colab. This allows users to:
- Access files stored in Google Drive
- Work with large datasets
- Save notebooks online

Installing Packages:
I also learned on how to install additional Python libraries in Colab using:
!pip install package_name

Machine Learning Basics:
I got a basic introduction to machine learning workflows in Colab. I learned about:
- Loading datasets
- Training models
- Testing models
- Predicting outputs

Advantages of Google Colab:
I understood the major advantages of Google Colab:
- Free to use
- Beginner friendly
- Cloud-based platform
- Supports collaboration
- Provides free GPU support
- Useful for AI and machine learning projects

Overall, Google Colab helped me understand how Python programming can be used for data analysis, visualization, and machine learning in a simple and practical way. Working with datasets, creating graphs, and using Python libraries improved my technical knowledge and gave me a better understanding of real-world applications of programming. The hands-on examples made the concepts easier to understand and increased my interest in exploring more topics related to data science and artificial intelligence.

This experience also improved my confidence in using cloud-based coding platforms and experimenting with different tools and libraries. I understood how useful Google Colab is for students, beginners, and researchers because it provides an easy and powerful environment for coding without requiring high-end systems or complicated installations. I believe the knowledge gained through this practice will help me in academic projects, practical assignments, and future learning in Python, machine learning, and AI-related technologies.